Two Hidden Layers Suffice: The Caterpillar Net

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Abstract

Every continuous piecewise linear (CPWL) function is a finite signed sum of maxima of affine functions, making maximum representation a central problem in the study of the expressiveness of ReLU networks. Earlier work disproved the conjectured base-two depth bound, leaving open whether a constant number of hidden layers suffices to represent the maximum. Independently and concurrently, Brunck, Hertrich, Rueß, Stade, and Yehudayoff established that two hidden layers suffice for all CPWL functions. We give the *Caterpillar Net*, a non-recursive explicit construction representing the maximum of N real inputs exactly with two hidden layers of widths $\binom{N}{2} + N + 1$ and $2^{O(N \log N)}$, rational weights, and zero biases. Every expansion of the maximum in its finite vocabulary compiles with the same first hidden layer, and the vocabulary has dimension $2^{\Theta(N \log N)}$ modulo its auxiliary functions. Sharing these layers across a Wang–Sun representation gives two-hidden-layer networks for all CPWL functions. [1]

1 Introduction

In this paper we investigate the representation capabilities of ReLU networks: what functions are representable by such networks of different depths and widths? ReLU networks represent precisely the continuous piecewise-linear (CPWL) functions with finitely many pieces [ABMM18]. A sharp question is therefore: given a CPWL function, what is the smallest depth, or the smallest width at a prescribed depth, needed to represent it exactly [HBD+23, Saf26]? The maximum is a universal building block for such functions: every CPWL function is a finite signed sum of maxima of affine functions [WS05]. Write

$$\max_N(x) := \max_{0 \le i < N} x_i. \tag{1}$$

Theorem (Wang–Sun, [WS05], `Lean_1`)**.** *Let $d \ge 1$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous and affine on each member of a finite polyhedral cover. There are $R \in \mathbb{N}$, affine maps and signs*

$$A_t : \mathbb{R}^d \longrightarrow \mathbb{R}^{d+1}, \qquad \varepsilon_t \in \{-1, 1\} \qquad (1 \le t \le R) \tag{2}$$

such that

$$f(x) = \sum_{t=1}^{R} \varepsilon_t \max_{d+1}(A_t(x)) \qquad \textit{for every } x \in \mathbb{R}^d. \tag{3}$$

[1] Our results were obtained over months of proof search in Lean using Claude Code and Codex; the AI methodology statement describes their roles. The Lean code is available in the accompanying repository https://github.com/JayPiZimmermann/Max_Caterpillar. Links to the Lean formalization appear in theorem titles and equation numbers; equation numbers are clickable.

Thus a two-hidden-layer representation of $\max_{d+1}$ gives one for every CPWL function on $\mathbb{R}^d$: precompose with the affine maps and add the networks in parallel. Section 7 shares parts of these parallel networks; the representation theorem and the resulting network theorems are formalized.

Our contribution. The *Caterpillar Net* gives a nonrecursive two-hidden-layer representation of the maximum with explicit bounds on both hidden-layer widths. The construction uses a finite dictionary of support functions of caterpillar polytopes and their faces. The maximum is the support function of a simplex. We first express this support function as an alternating sum of caterpillar support functions, up to a linear combination of support functions of lower-dimensional polytopes. We then represent the remaining error using faces of the same caterpillar polytopes and elementary ReLU features. Normalizing and merging these descriptions yields a width bound for the complete network, including all correction terms.

Write $\sigma(t) := \max(0, t)$ applied coordinatewise to vectors. A network has an input layer of N coordinates, first and second hidden layers of widths h_1 and h_2, respectively, and a scalar output layer. With real weight matrices W_1, W_2, output vector w_3, and biases b_1, b_2, b_3 of the corresponding dimensions, it evaluates as

$$x \longmapsto w_3^\top \sigma\big(W_2\sigma(W_1x + b_1) + b_2\big) + b_3. \tag{4}$$

There are no skip connections; equality is required on all of $\mathbb{R}^N$. For $N = 1, 2, 3$ there are known simple network representations of width $(2, 2)$ and $(4, 2)$ and $(7, 5)$. Our Main Theorem presents such constructions for higher N. A network has *rational weights* if every entry of W_1, W_2, and w_3 is rational, and *zero biases* if b_1, b_2, and b_3 vanish.

Theorem 1 (Caterpillar Net, `Lean_2`, `Lean_3`, Proof)**.** *For every $N \geq 3$, there is a two-hidden-layer ReLU network with rational weights and zero biases computing $\max_N$ with widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = 3^{N-2}C_N - N(N-1) + 2, \tag{5}$$

where

$$C_N := \sum_{m=2}^{N-1} \binom{N}{m} \frac{m!}{2}(N-m)m^{N-m-1}. \tag{6}$$

Thus $h_1 = O(N^2)$ and $h_2 = 2^{O(N \log N)}$ as $N \to \infty$.

Further results. The network is compiled from an expansion of $\max_N$ in a finite vocabulary of caterpillar face functions. Theorem 7 separates the network from the coefficients: any such expansion with $|K|$ retained units compiles with the same first hidden layer and $|K| + 2$ second hidden layer neurons. Rational coefficients exist without any integration (Appendix A), and the canonical coefficients behind Theorem 1 are rational as well. Section 6 bounds the second width by $q_N + 2$, where q_N is the dimension of the vocabulary modulo its auxiliary functions, and proves $q_N \geq (N-2)!$; hence $\log q_N = \Theta(N \log N)$. For $N = 3, 4$ an exact search finds expansions with one and three units. Section 7 applies the construction to sums of maxima.

2 Related work

The conjecture that $\max\{0, x_1, \ldots, x_{2^k}\}$ requires more than k hidden layers [HBD+23] was disproved for $k = 2$ by the maximum of five inputs in [BBH+26]. Exact computer-assisted searches subsequently obtained

two-hidden-layer representations up to $N = 12$ in [RAB+26] and, independently, up to $N = 8$ in [WB26]. The improved general depth bounds remained logarithmic, leaving constant depth open.

Depth lower bounds hold for integer weights [HHL23] and rational weights whose denominators are restricted to powers of a fixed integer [AHM25]. Our weights are rational, but their denominators are not restricted in this way, so neither bound applies. Other results concern braid-fan-compatible networks [GHL25] and first hidden layer width bounds [Saf26, Lu26].

Theorem 2 (Brunck, Hertrich, Rueß, Stade, and Yehudayoff, [BHR+26])**.** *Every continuous piecewise affine function $f : \mathbb{R}^d \to \mathbb{R}$ with finitely many affine pieces can be represented exactly by a ReLU network with two hidden layers.*

Their first proof proceeds by induction on dimension, using homological algebra to construct a linear combination of support functions of pyramids over parallelepipeds that agrees with the simplex support function modulo lower-dimensional terms. Our construction restricts these building blocks to an explicitly specified finite family, the caterpillar polytopes; we deliver a geometric proof of the corresponding identity.

Independently and concurrently, they developed a second proof based on combinatorial cancellation, using the same caterpillar construction. The caterpillar family and coefficients coincide with ours. We establish explicit width bounds for the first and second hidden layers, respectively. These bounds also apply to their combinatorial construction but are not stated in their paper.

3 The Caterpillar Net

Two identities drive the construction:

$$\max(s, t) = t + \sigma(s - t), \tag{7}$$

$$\sigma(\ell(x)) = \sigma(-\ell(x)) + \ell(x). \tag{8}$$

The first suggests a path with attached leaves. A signed sum of the resulting caterpillar support functions agrees with the maximum modulo lower-dimensional terms; proper faces and linear functions correct the error. The second identity reverses hinges to merge units in Section 5.

Fix $N \geq 3$ and write $x = (x_0, \ldots, x_{N-1}) \in \mathbb{R}^N$.

Definition 3 (Labelled caterpillar, `Lean_4`)**.** A *caterpillar description* is a permutation–set pair

$$\pi \in \mathrm{Sym}\{0, \ldots, N-1\}, \qquad \varnothing \neq S \subseteq \{1, \ldots, r\}, \qquad r := N - 2. \tag{9}$$

Position j carries label $\pi(j+1)$, and $\pi(0)$ is the *apex*. The *spine positions* are $\{0\} \cup S$. The *cut set* S separates positions $j-1$ and j whenever $j \in S$. The last spine position q and the *parent* $p(j)$, the nearest preceding spine position, are

$$q := \max S, \qquad p(j) := \max\big(\{0\} \cup \{i \in S : i < j\}\big) \qquad (1 \leq j \leq r). \tag{10}$$

The parent–child edges form the *labelled caterpillar*: a path through the spine positions with each remaining label attached as a leaf to its parent (Figure 1). The apex is separate from the tree.

Definition 4 (Caterpillar function and face functions, `Lean_5`, `Lean_6`)**.** For labels i, j, for the tree edges, and for the head, put

$$\ell_{ij}(x) := x_i - x_j, \tag{11}$$

$$L_{\pi,S} := \big\{\ell_{\pi(p(j)+1),\pi(j+1)} : 1 \leq j \leq r\big\}, \tag{12}$$

$$h_{\pi,S} := \ell_{\pi(q+1),\pi(0)}. \tag{13}$$

The *edge functions* $L_{\pi,S}$ subtract each child's coordinate from its parent's, one for each of the r tree edges, and the *head function* $h_{\pi,S}$ compares the last spine vertex with the apex. For a linear function h, a finite set L of linear functions, and a *state* $\tau(\ell) \in \{0,\ \ell,\ \sigma \circ \ell\}$ chosen for every $\ell \in L$, define

$$U^{\tau}_{h,L}(x) := \sigma\Big(h(x) + \sum_{\ell \in L} \tau(\ell)(x)\Big). \tag{14}$$

Each edge is omitted, kept linearly, or kept as a hinge. Write $U_{h,L}$ for the all-hinge state, and call (h, L) its *function description*, recording the head and inner hinges. The *caterpillar function* of (π, S) and its *face functions* are

$$U_{\pi,S} := U_{h_{\pi,S},L_{\pi,S}}, \qquad U^{\tau}_{\pi,S} := U^{\tau}_{h_{\pi,S},L_{\pi,S}}. \tag{15}$$

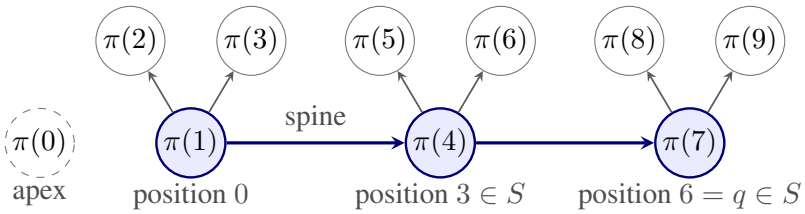


Figure 1: A labelled caterpillar with $N = 10$ and $S = \{3, 6\}$. The spine $\pi(1) \to \pi(4) \to \pi(7)$ has two leaves at each vertex; all arrows point from parent to child. The apex $\pi(0)$ is separate from the tree. The last spine position is $q = 6$, so the head function is $h_{\pi,S}(x) = x_{\pi(7)} - x_{\pi(0)}$.

The term *face function* comes from the following geometry. Identify $a(x) = \langle a, x\rangle$ with its coefficient vector. Then $\ell_{ij} = e_i - e_j$ points from e_j to e_i, and $h = h_{\pi,S} = e_{\pi(q+1)} - e_{\pi(0)}$ points from the apex label to the last spine label. Each $\ell \in L = L_{\pi,S}$ points from child to parent. As free vectors, h places a base corner and the ℓ generate its edges. The polytope apex is 0, distinct from the labelled point $e_{\pi(0)}$. Define the *support function* and the caterpillar polytope by

$$h_P(x) := \max_{p \in P} \langle p, x\rangle, \tag{16}$$

$$Q_{h,L} := \operatorname{conv}\Big(\{0\} \cup \Big(h + \sum_{\ell \in L} [0, \ell]\Big)\Big), \qquad U_{h,L} = h_{Q_{h,L}}. \tag{17}$$

Indeed, the segment $[0, \ell]$ has support function $\sigma \circ \ell$, Minkowski sums add support functions, and adjoining 0 applies the outer ReLU. The base is the translated zonotope $h + \sum_{\ell \in L}[0, \ell]$. For a state τ in Equation (14), set $t_\ell = \mathbb{1}_{\{\tau(\ell) = \ell\}}$, where $\mathbb{1}$ denotes the indicator. It is zero for omitted edges and hinges; the latter contribute segments. The corresponding polytope is $Q^{\tau}_{h,L} := \operatorname{conv}\big(\{0\} \cup (h + \sum_{\ell \in L} t_\ell \ell + \sum_{\tau(\ell) = \sigma \circ \ell}[0, \ell])\big)$, and $U^{\tau}_{h,L} = h_{Q^{\tau}_{h,L}}$. Figure 3 illustrates the translated generators for $N = 4$ on the slice $x_0 = 0$, where a becomes (a_1, a_2, a_3) and e_0 is represented by 0.

The head connects the apex label to the tree, so the head and $N-2$ edge functions are linearly independent. Thus the base is an $(N-2)$-dimensional zonotope with affine hull avoiding 0, and $Q_{h,L}$ is an $(N-1)$-dimensional pyramid. Each nontrivial face containing 0 cones over a base face. Maximizing on each base segment selects 0, ℓ, or $[0, \ell]$, with support functions 0, ℓ, and $\sigma \circ \ell$; the outer ReLU adjoins the apex. These are precisely the three states in Equation (14).

Definition 5 (Dictionary, `Lean_7`)**.** The *dictionary* $\mathcal{D}$ is the finite set of functions $\mathbb{R}^N \to \mathbb{R}$ consisting of every face function $U^\tau_{\pi,S}$, over all caterpillar descriptions (π, S) and all states τ, together with the *auxiliary functions*

$$\begin{aligned} &x \mapsto \sigma(x_i), \quad x \mapsto \sigma(-x_i) \quad (0 \le i < N), \\ &x \mapsto \sigma(x_i - x_j) \qquad\qquad\quad (0 \le i < j < N). \end{aligned} \tag{18}$$

Heads and edge functions are coordinate differences, so face functions and the functions $x \mapsto \sigma(x_i - x_j)$ are translation invariant: with $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^N$,

$$U^\tau_{\pi,S}(x + t\mathbf{1}) = U^\tau_{\pi,S}(x) \qquad (t \in \mathbb{R}). \tag{19}$$

Only $\sigma(\pm x_i)$ change under this translation. Theorem 8 proves that $\mathcal{D}$ spans the maximum. Appendix A gives nonrecursive coefficients by averaging inverse-Gram expansions over its bases:

Lemma 6 (Canonical expansion, `Lean_8`, Proof)**.** *Assuming Theorem 8, the coefficients Equation* (89) *satisfy*

$$\max_N = \sum_{g \in \mathcal{D}} c_g g. \tag{20}$$

3.1 The network

The network computes an expansion of $\max_N$ in face functions and auxiliary functions. To compile face functions, first absorb every linear edge of a face state into the head:

$$U^\tau_{h,L} = U_{h',M}, \qquad h' := h + \sum_{\tau(\ell)=\ell} \ell, \qquad M := \{\ell \in L : \tau(\ell) = \sigma \circ \ell\}. \tag{21}$$

By Equation (8), reversing each remaining ℓ_{ij} with $i > j$ gives the same function with the *normalized description*

$$\begin{aligned} M^+ &:= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in M\}, \\ h^+ &:= h' + \sum_{\ell_{ij} \in M,\ i>j} \ell_{ij}. \end{aligned} \tag{22}$$

Geometrically, $[0, \ell] = \ell + [0, -\ell]$: reversing a generator shifts the head by the old generator. Normalization chooses a consistent orientation and a corresponding base corner; it does not rescale vectors or change the polytope (Figure 3). Write $U_k := U_{h,M}$ for a normalized description $k = (h, M)$, and let $\mathcal{K}_0$ be the set of normalized descriptions of all face functions. Let A_N be the span of the auxiliary functions Equation (18). The network has three parts (Figure 2).

- The *first hidden layer* computes the shared features

$$\Phi(x) := \big(\sigma(x_0),\ \sigma(-x_0),\ (\sigma(x_i - x_0))_{i>0},\ (\sigma(x_i - x_j))_{i<j}\big). \tag{23}$$

- The *second hidden layer* computes the retained units U_k, $k \in K$, for a set $K \subseteq \mathcal{K}_0$, and the two neurons $\sigma(R)$ and $\sigma(-R)$ for an auxiliary remainder $R \in A_N$.
- The *output layer* returns $\sum_{k \in K} d_k U_k + \sigma(R) - \sigma(-R)$.

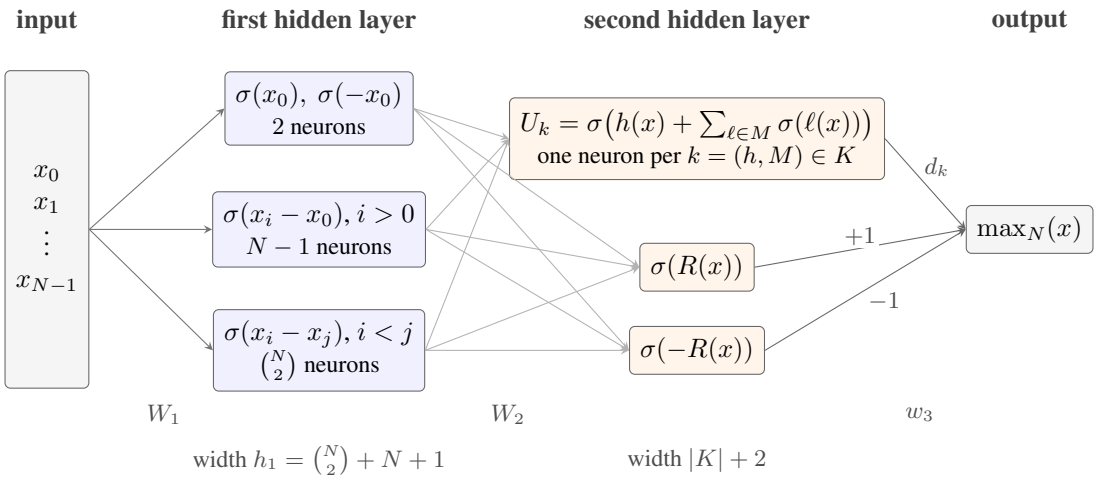


Figure 2: The Caterpillar Net. The first hidden layer is fixed. The second hidden layer has one neuron for every retained normalized description $k = (h, M)$ and two neurons for the auxiliary remainder R. All biases are zero; the matrices are given in Equation (30).

Theorem 7 (Compilation, `Lean_9`, Proof)**.** *Let $K \subseteq \mathcal{K}_0$, $d \in \mathbb{R}^K$, and $R \in A_N$ satisfy*

$$\max_N = \sum_{k \in K} d_k U_k + R. \tag{24}$$

Then the network with first hidden layer Equation (23) *and the matrices in Equation* (30) *computes* $\max_N$*. It has widths h_1 and $|K| + 2$ and zero biases. Its weights are rational whenever the coefficients d_k and the auxiliary coefficients of R are rational.*

Proof of Theorem 7. The first hidden layer outputs Φ from Equation (23). Thus W_1 has rows $e_0^\top$ and $-e_0^\top$, then $e_i^\top - e_0^\top$ for $i > 0$, then $e_i^\top - e_j^\top$ for $i < j$, using a fixed ordering within each block. Its row count is

$$\binom{N}{2} + (N-1) + 2 = \binom{N}{2} + N + 1 = h_1. \tag{25}$$

By Equation (8), these features recover all coordinates,

$$\begin{aligned} x_0 &= \sigma(x_0) - \sigma(-x_0), \\ x_i &= x_0 + \sigma(x_i - x_0) - \sigma(x_0 - x_i) \qquad (i > 0). \end{aligned} \tag{26}$$

For each $k = (h, M) \in K$, substitute Equation (26) into $h(x) + \sum_{\ell \in M} \sigma(\ell(x))$ and collect the coefficients of $\Phi(x)$ into a row vector $w_k^\top$. Every hinge in M already has the orientation $i < j$, so its coefficient is added directly to the corresponding feature.

To specify the remaining row, collect the auxiliary coefficients of R as

$$R(x) = \sum_i p_i \sigma(x_i) + \sum_i n_i \sigma(-x_i) + \sum_{i<j} s_{ij} \sigma(x_i - x_j). \tag{27}$$

Each retained unit is translation invariant by Equation (19), and $\max_N(x + t\mathbf{1}) = \max_N(x) + t$, so Equation (24) gives

$$R(x + t\mathbf{1}) = R(x) + t, \qquad \sum_i p_i = 1. \tag{28}$$

The coefficient sum follows by setting $x = 0$ and $t = 1$. Now choose t large enough that every $x_i + t$ is nonnegative. Substituting the auxiliary expansion into Equation (28) cancels t and gives

$$R(x) = \sum_i p_i x_i + \sum_{i<j} s_{ij}\sigma(x_i - x_j). \tag{29}$$

Substitute Equation (26) into this expression and collect the coefficients of $\Phi(x)$ into w_R. Thus both w_k and w_R are specified by finite sums of the previously defined coefficients. The entries of W_1 and of the rows w_k are integers, because heads and hinges are integer combinations of coordinates. The row w_R is a rational combination of the p_i and s_{ij}, and w_3 consists of the d_k and ± 1.

Order K and set the second hidden layer matrix and output layer vector to

$$W_2 = \begin{pmatrix} (w_k^\top)_{k\in K} \\ w_R^\top \\ -w_R^\top \end{pmatrix}, \qquad w_3 = \begin{pmatrix} (d_k)_{k\in K} \\ 1 \\ -1 \end{pmatrix}, \qquad b_1 = b_2 = b_3 = 0. \tag{30}$$

The second hidden layer outputs $U_k(x)$ and $\sigma(\pm R(x))$. By Equation (8) and Equation (24),

$$w_3^\top \sigma(W_2\Phi(x)) = \sum_{k\in K} d_k U_k(x) + \sigma(R(x)) - \sigma(-R(x)) = \max_N(x). \tag{31}$$

□

The coefficients of Lemma 6 give such an expansion; Section 5 counts the retained descriptions.

4 Why the dictionary spans the maximum

The maximum is the support function of a simplex. To work in its affine dimension, set

$$\Delta := \operatorname{conv}\{0, e_1, \ldots, e_{N-1}\} \subseteq H_0 := \{x \in \mathbb{R}^N : x_0 = 0\}. \tag{32}$$

For every $x \in \mathbb{R}^N$,

$$\max_N(x) = x_0 + h_\Delta(x - x_0\mathbf{1}). \tag{33}$$

It suffices to prove that h_Δ lies in the span of the restrictions of $\mathcal{D}$ to H_0: the substitution $x \mapsto x - x_0\mathbf{1}$ fixes every face function, sends coordinate features to $\sigma(\pm(x_i - x_0))$, and preserves the dictionary span by Equation (8).

Henceforth all polytopes lie in H_0, and heads and edge functions are identified with their restricted coefficient vectors. Let $\mathcal{L}_k$ be the real span of support functions of polytopes of affine dimension at most k; thus $\mathcal{L}_0$ consists of linear functions. The *exposed face* of P in direction $u \in H_0$ is

$$P^u := \{p \in P : \langle p, u\rangle = h_P(u)\}. \tag{34}$$

The opposite face P^{-u} consists of the minimizers. Define

$$D_u h_P(x) := h_{P^u}(x) - h_{P^{-u}}(x) \qquad (x \in H_0). \tag{35}$$

Here u selects the faces, while x is the support-function input; $P^0 = P$ gives $D_0 = 0$. For a simplex facet normal, D_u subtracts the opposite vertex's linear support function from that of the facet.

For a finite linear combination g of support functions, define its one-sided directional derivative at u by

$$\begin{aligned} \partial_u g(x) &:= \lim_{t\downarrow 0} \frac{g(u+tx)-g(u)}{t}, \\ \partial_u h_P = h_{P^u}, &\qquad D_u g = \partial_u g - \partial_{-u} g. \end{aligned} \tag{36}$$

This makes D_u independent of the support-function decomposition. When $\langle u, \cdot\rangle$ is constant on P, the exposed faces cancel; otherwise both have smaller dimension. Consequently,

$$D_u \mathcal{L}_k \subseteq \mathcal{L}_{k-1} \qquad (k \geq 1). \tag{37}$$

The signed caterpillar sum. We seek a combination that agrees with h_Δ up to a function in $\mathcal{L}_{N-2}$. Adding $x_{\pi(0)}$ makes the caterpillar function the following maximum

$$x_{\pi(0)} + U_{\pi,S}(x) = \max\left\{ x_{\pi(0)},\, x_{\pi(q+1)} + \sum_{\ell \in L_{\pi,S}} \sigma(\ell(x)) \right\}. \tag{38}$$

Averaging over permutations gives the required symmetry

$$F_N(x) := \frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,r\}} (-1)^{N+|S|} \big(x_{\pi(0)} + U_{\pi,S}(x)\big), \tag{39}$$

The signs cancel the extra exposed-face contributions modulo $\mathcal{L}_{N-3}$, as proved below.

Theorem 8 (Dictionary sufficiency, `Lean_10`, Proof)**.** *For every $N \geq 3$, the dictionary of Definition 5 satisfies*

$$\max_N \in \operatorname{span} \mathcal{D}. \tag{40}$$

Proof of Theorem 8. Define the error by

$$f := F_N|_{H_0} - h_\Delta. \tag{41}$$

Step 8.1 reduces dictionary membership to $f \in \mathcal{L}_{N-2}$; Step 8.2 reduces this to $D_u f \in \mathcal{L}_{N-3}$ for all u. Step 8.3 proves these bounds by cancellation, using induction only when deleting a single exceptional coordinate.

Step 8.1 (Reduce to a lower-dimensional error). Assume temporarily that

$$f = F_N - h_\Delta \in \mathcal{L}_{N-2}. \tag{42}$$

Write $Q_{\pi,S} := Q_{h_{\pi,S}, L_{\pi,S}}$. Removing the linear apex terms from Equation (39) gives

$$\frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,N-2\}} (-1)^{N+|S|} h_{Q_{\pi,S}} - h_\Delta \in \mathcal{L}_{N-2}. \tag{43}$$

We correct the error by subtracting derivatives ∂_u until it is linear. Faces of $Q_{\pi,S}$ already have support functions in the dictionary span: apex–base faces give Equation (14), base faces give linear functions plus hinges, and the apex alone gives zero. Coordinate features and Equation (8) represent the linear terms and either hinge orientation. Differentiating h_Δ also produces simplex faces; induction on their vertex count supplies these functions.

Claim 8.1 [Simplex faces, `Lean_11`, `Lean_12`, Proof]. For every nonempty $T \subseteq \{0, e_1, \dots, e_{N-1}\}$, the support function of $\Delta_T := \operatorname{conv} T$ lies in the dictionary span.

Start ($|T| = 1$). The support function of a singleton is linear and lies in the dictionary span.

Step ($|T| \geq 2$). Fix T and assume the claim for all smaller nonempty vertex sets. For any current vertex set A with $|A| \geq 2$, remove $v \in A$ using a direction whose value at v exceeds its common value at the other vertices. Writing $\Delta_A := \operatorname{conv} A$, this gives

$$\begin{aligned} u_v &:= \begin{cases} -\sum_{i=1}^{N-1} e_i & v = 0, \\ v & v \neq 0, \end{cases} \\ (\Delta_A)^{u_v} &= \{v\}, \qquad (\Delta_A)^{-u_v} = \Delta_{A\setminus\{v\}}, \\ D_{u_v} h_{\Delta_A}(x) &= \langle v, x\rangle - h_{\Delta_{A\setminus\{v\}}}(x). \end{aligned} \tag{44}$$

Order the vertices outside T as $v_1, \ldots, v_{N-|T|}$ and set

$$\varepsilon_T := D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} \left(\frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\ldots,N-2\}} (-1)^{N+|S|} h_{Q_{\pi,S}} - h_\Delta \right). \tag{45}$$

Each deletion changes the sign of the remaining simplex term, and subsequent D_u annihilate the linear terms. Hence

$$D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} h_\Delta = (-1)^{N-|T|} h_{\Delta_T} + \ell_T, \tag{46}$$

where ℓ_T is linear. Thus the coefficient of h_{Δ_T} in ε_T is $(-1)^{N-|T|+1}$; all other terms are linear or caterpillar-face support functions and lie in $\operatorname{span} \mathcal{D}$. By Equation (37), $\varepsilon_T \in \mathcal{L}_{|T|-2}$. For the full vertex set the empty composition is the identity.

Lower the residual dimension recursively by

$$\begin{aligned} \varepsilon_T^{(0)}(x) &:= \varepsilon_T(x), \\ \varepsilon_T^{(j+1)}(x) &:= \varepsilon_T^{(j)}(x) - \sum_{u \in E_{T,j}} \big(\partial_u \varepsilon_T^{(j)}\big)(x). \end{aligned} \tag{47}$$

Here $0 \leq j < |T| - 2$; the next claim chooses each finite $E_{T,j} \subseteq H_0$.

Claim 8.2 [Residual reduction, `Lean_13`, Proof]. The sets $E_{T,j}$ can be chosen successively so that, for $0 \leq j \leq |T| - 2$,

$$\begin{aligned} \varepsilon_T^{(j)} &\in \mathcal{L}_{|T|-2-j}, \\ \varepsilon_T - \varepsilon_T^{(j)} &\in \operatorname{span} \mathcal{D}. \end{aligned} \tag{48}$$

Each residual remains a linear combination of h_{Δ_T}, support functions of its proper faces and of caterpillar faces, and linear functions.

The proof in Appendix B.1 chooses a direction orthogonal to each maximal-dimensional direction space, avoiding orthogonality to the simplex and other spaces. The derivative sum preserves each maximal term exactly once; subtracting it lowers the dimension.

At stage $j = |T| - 2$, the error is linear, so Equation (48) gives $\varepsilon_T \in \operatorname{span} \mathcal{D}$. Its simplex coefficient is ± 1, and its other terms already lie in the span; hence $h_{\Delta_T} \in \operatorname{span} \mathcal{D}$. Taking the full vertex set proves $h_\Delta \in \operatorname{span} \mathcal{D}$ under assumption Equation (42).

Step 8.2 (Why the facet bounds imply the class identity). For $N \geq 4$ and $f = F_N - h_\Delta$, we show that $f \in \mathcal{L}_{N-2}$ follows from

$$D_u f \in \mathcal{L}_{N-3} \qquad \text{for every } u \in H_0. \tag{49}$$

The base zonotope admits the Freudenthal–Kuhn triangulation [Fre42, Kuh60]; coning it to the apex expresses each caterpillar support function as a sum of simplex support functions modulo $\mathcal{L}_{N-2}$. For a simplex, D_u isolates a facet up to the linear support function of its opposite vertex. Thus Equation (49) controls the weighted sums of parallel facets. A common linear reconstruction operator turns these facet cancellations back into a cancellation of the original simplex sum. Define

$$(\mathcal{T}g)(x) := \begin{cases} g(x) & x_1 \geq 0, \\ g(x - x_1 e_1) & x_1 < 0. \end{cases} \tag{50}$$

Claim 8.3 [Simplex reconstruction, `Lean_14`, Proof]. Every $(N-1)$-dimensional simplex $P \subseteq H_0$ satisfies

$$h_P - \mathcal{T}\left(\sum_{F \text{ facet of } P} \operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F \right) \in \mathcal{L}_{N-2}. \tag{51}$$

Here ν_F is an outward normal of F. Its sign against e_1 records whether the facet bounds lines parallel to e_1 from above or below; parallel facets contribute zero. The operator $\mathcal{T}$ fills the segments between a polytope and its projection onto $p_1 = 0$, up to a segment correction for translation; hence $\mathcal{T}\mathcal{L}_k \subseteq \mathcal{L}_{k+1}$. Applying the claim with the same operator to every simplex preserves the cancellations between parallel facets. Their sums lie in $\mathcal{L}_{N-3}$, so their reconstructions lie in $\mathcal{L}_{N-2}$, as required. Appendix B.2 proves the claim and the subdivision and grouping steps.

Claim 8.4 [Class identity, `Lean_15`, Proof]. The class identity Equation (42) holds for every $N \geq 3$.

Step 8.3 (Prove the class identity by induction).

Start ($N = 3$). Here $S = \{1\}$, $q = 1$, and $p(1) = 0$, so Equation (38) gives

$$\begin{aligned} x_{\pi(0)} + U_{\pi,\{1\}}(x) &= \max\bigl\{x_{\pi(0)}, x_{\pi(2)} + \sigma(x_{\pi(1)} - x_{\pi(2)})\bigr\} \\ &= \max\{x_{\pi(0)}, x_{\pi(1)}, x_{\pi(2)}\} = \max_3(x). \end{aligned} \tag{52}$$

All six terms have coefficient $1/6$, so

$$F_3 = \max_3 . \tag{53}$$

On H_0, this gives $f = 0 \in \mathcal{L}_1$.

Step ($N \geq 4$). Assume the class identity for $N-1$. By Step 8.2, it suffices to prove $D_u f \in \mathcal{L}_{N-3}$ for every $u \in H_0$. The linear apex terms disappear. The exposed simplex faces have the largest and smallest coordinate values u_i, including $u_0 = 0$. In a caterpillar base, $[0, \ell_{ij}]$ remains free when $u_i = u_j$; its maximizing endpoint is ℓ_{ij} when $u_i > u_j$ and 0 when $u_i < u_j$. Minimization reverses these choices.

Discarding contributions in $\mathcal{L}_{N-3}$. Fix $u \in H_0 \setminus \{0\}$, since $D_0 f = 0$, and a summand $Q_{\pi,S}$. A face lying entirely in its base $B_{\pi,S} := h_{\pi,S} + \sum_{\ell \in L_{\pi,S}} [0, \ell]$, without the apex 0, is a translated sum of segments; its support function lies in $\mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. The singleton apex has zero support function. If at least two tree edges have endpoint labels i, j with $u_i \neq u_j$, at most $N-4$ segments remain free; adjoining the apex gives dimension at most $N-3$. Only apex–base faces with at most one tree edge whose endpoint coordinate values differ remain.

For a face exposed in direction $-u$, replace u by $-u$; we can then always analyze maximizers of $\langle u, \cdot \rangle$. Put $a := u_{\pi(0)}$. The face contains both the apex and base points exactly when the maximum over the base is zero. There are three cases.

Case 1: one spine edge with unequal endpoint values. Removing this edge splits the tree into two components with constant coordinate values $c \neq d$. The maximum over the base is $\max\{c, d\} - a$, so the required condition is

$$a = \max\{c, d\}. \tag{54}$$

The apex label therefore adds no new coordinate value, and u has exactly two distinct coordinate values.

Case 2: one leaf edge with unequal endpoint values. Let d be the leaf's coordinate value and $c \neq d$ the common value at all other tree vertices. The maximum over the base is $c + \sigma(c-d) - a$, so the required condition is

$$a = c + \sigma(c-d) = \begin{cases} c, & d > c, \\ 2c - d, & d < c. \end{cases} \tag{55}$$

If $d > c$, the apex label has value c, giving exactly two distinct coordinate values. If $d < c$, put $t := c - d > 0$. The leaf has value $c - t$, the apex label has value $c + t$, and all other $N-2$ labels have value c. Up to a permutation of labels, this gives the three-value pattern $(c + t, c - t, c, \ldots, c)$.

Case 3: equal values at every tree-edge endpoint pair. If $u_i = u_j$ for the endpoint labels of every tree edge, all tree values equal some c. The maximum over the base is $c - a$, so we would also need $a = c$, making every coordinate of u equal. Since $u \in H_0$, this would force $u = 0$, which we excluded.

Negating u preserves these patterns. Since $Q_{\pi,S}$ was arbitrary, only directions with two distinct coordinate values or the pattern $(c + t, c - t, c, \ldots, c)$ can contribute to the full signed sum.

Cancellation for three distinct values. Let α, β be the exceptional labels, with $u_\alpha = c + t$, $u_\beta = c - t$, and $u_i = c$ for every other label, where $t \neq 0$. Let τ exchange α and β and fix all other labels. Define the linear map $R : H_0 \to H_0$ explicitly by

$$(Rx)_i := x_{\tau(i)} - x_{\tau(0)} \qquad (0 \leq i < N,\ x \in H_0). \tag{56}$$

Thus R first swaps the two exceptional coordinates and then subtracts the new zeroth coordinate from every coordinate; in particular, $(Rx)_0 = 0$. Since $u_0 = 0$, this formula also gives $Rx = x - \frac{x_\alpha - x_\beta}{t} u$. With the explicit linear functional $\langle b, x \rangle := (x_\alpha - x_\beta)/(2t)$, we therefore have $Rx = x - 2\langle b, x \rangle u$, $\langle b, u \rangle = 1$, and $Ru = -u$. Hence R is a rank-one reflection. Permutation symmetry and the common coordinate shift $-x_{\tau(0)}$ give $F_N(Rx) = F_N(x) + \ell(x)$ for a linear function ℓ. Write $G := F_N|_{H_0}$. Applying the reflection identity to $u + \varepsilon x$ and to u, subtracting, dividing by $\varepsilon > 0$, and taking $\varepsilon \downarrow 0$ gives $\partial_{-u} G(Rx) = \partial_u G(x) + \ell(x)$, because $R(u + \varepsilon x) = -u + \varepsilon Rx$. To remove R from the derivative argument, positive homogeneity gives

the identity $\partial_v G(x+\lambda v) = \partial_v G(x) + \lambda G(v)$ for every real λ. Indeed, for sufficiently small $\varepsilon > 0$, write $v + \varepsilon(x+\lambda v) = (1+\varepsilon\lambda)\big(v + \frac{\varepsilon}{1+\varepsilon\lambda}x\big)$ and apply the definition Equation (36). Using this identity with $v = -u$ and $\lambda = 2\langle b, x\rangle$, we obtain

$$\begin{aligned} \partial_u G(x) + \ell(x) &= \partial_{-u} G(Rx) \\ &= \partial_{-u} G\big(x + 2\langle b, x\rangle(-u)\big) \\ &= \partial_{-u} G(x) + 2\langle b, x\rangle G(-u), \\ D_u G(x) &= \partial_u G(x) - \partial_{-u} G(x) = 2F_N(-u)\langle b, x\rangle - \ell(x). \end{aligned} \tag{57}$$

The right-hand side is linear in x, so

$$D_u(F_N|_{H_0}) \in \mathcal{L}_0. \tag{58}$$

The two exposed simplex faces are vertices, so their support functions are linear as well.

Handling two distinct values. First suppose that each value occurs at least twice. After the preceding dimension bound, every term still requiring cancellation has a unique split $u_{\pi(1)} = \cdots = u_{\pi(k)} \neq u_{\pi(k+1)} = \cdots = u_{\pi(r+1)}$ with $k \in S$. If the apex label has the latter value, toggle membership of $k-1$ in S. Otherwise, exchange the equal-valued labels $\pi(k+1), \pi(k+2)$ and toggle membership of $k+1$. Repetition of both values gives $k \geq 2$ in the first case and $k+2 \leq r+1$ in the second, so the operations are defined; the unchanged cut at k keeps the cut set nonempty. The transformed term remains in the same case, and repeating the operation returns the original term. If Φ denotes this involution on the active terms and $w_{\pi,S} := (-1)^{N+|S|}/N!$, then

$$w_{\Phi(\pi,S)} = -w_{\pi,S}, \qquad D_u h_{Q_{\pi,S}} - D_u h_{Q_{\Phi(\pi,S)}} \in \mathcal{L}_1. \tag{59}$$

Inactive fixed points of the global transformation are already in $\mathcal{L}_{N-3}$ by the dimension bound. Summing the active pairs therefore puts $D_u F_N$ in $\mathcal{L}_{N-3}$. Each exposed simplex face has at most $N-2$ vertices and hence dimension at most $N-3$, so $D_u f \in \mathcal{L}_{N-3}$ without using induction.

One exceptional coordinate: the induction hypothesis. Suppose one of the two values occurs only once. Choose τ with $\tau(0)$ the exceptional label. To stay in H_0, relabel and subtract the new zeroth coordinate:

$$B_\tau : H_0 \longrightarrow H_0, \qquad (B_\tau x)_i := x_{\tau(i)} - x_{\tau(0)} \qquad (0 \leq i < N). \tag{60}$$

Its inverse is $B_{\tau^{-1}}$, and it sends u to $(0, t, \ldots, t)$, $t \neq 0$. Transforming both input and direction preserves every $\mathcal{L}_k$ and changes f only by a linear term, annihilated by D. Rescaling, and reversing the sign of D_u if needed, reduces to $u = (0, 1, \ldots, 1)$.

This direction exposes the $(N-2)$-dimensional facet $\Delta' := \mathrm{conv}\{e_1, \ldots, e_{N-1}\}$ and the opposite vertex 0. Hence

$$D_u h_\Delta(x) = h_{\Delta'}(x) = \max\{x_1, \ldots, x_{N-1}\} \qquad (x \in H_0). \tag{61}$$

Here is the exact use of the induction hypothesis. Set $z := (x_1, \ldots, x_{N-1})$ and $y := z - x_1\mathbf{1} = (0, x_2 - x_1, \ldots, x_{N-1} - x_1)$, where $\mathbf{1} \in \mathbb{R}^{N-1}$ has every coordinate equal to 1. Every caterpillar maximum increases by a common added constant. Averaging over the $(N-1)!$ lower label orders cancels the factorial in the total signed coefficient, giving

$$F_{N-1}(z) - \max_{N-1}(z) = F_{N-1}(y) - \max_{N-1}(y) + (W_{N-1} - 1)x_1 \tag{62}$$

for

$$W_{N-1} := \sum_{\varnothing \neq T \subseteq \{1,\dots,N-3\}} (-1)^{N-1+|T|}. \tag{63}$$

Unlike z, the vector y has zeroth coordinate zero, so the $(N-1)$-input induction hypothesis applies to it. Linear precomposition preserves $\mathcal{L}_{N-3}$, and the last term is linear. Thus

$$\big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big) \in \mathcal{L}_{N-3}. \tag{64}$$

Claim 8.5 [Coordinate deletion, `Lean_16`, Proof]. For $N \geq 4$,

$$D_{(0,1,\dots,1)}(F_N|_{H_0}) - \big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1})\big) \in \mathcal{L}_{N-3}. \tag{65}$$

The proof is in Appendix B.3. Its mechanism is that only degree-one tree positions of label 0 survive. Each nonempty lower cut set has N insertion preimages, each contributing $1/N$ of its lower coefficient. The extra empty-cut star sum lies in $\mathcal{L}_{N-3}$ by Boolean cancellation.

Completion of the induction. Combining Equation (61), Equation (64), and Claim 8.5 gives

$$\begin{aligned} D_u f(x) = {} & \big[D_u(F_N|_{H_0})(x) - F_{N-1}(x_1, \dots, x_{N-1})\big] \\ & + \big[F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big]. \end{aligned} \tag{66}$$

Both brackets lie in $\mathcal{L}_{N-3}$, the first by Claim 8.5 and the second by induction. Step 8.2 now gives $f \in \mathcal{L}_{N-2}$, and Step 8.1 gives the dictionary representation. □

5 Network size

The widths in Theorem 1 are worst-case upper bounds: we count all possible descriptions after collecting linear terms into the head and orienting the remaining hinges consistently. Discarding zero aggregate coefficients can give a smaller network; zero padding realizes the stated widths.

The normalized description Equation (22) of a caterpillar function depends only on its undirected spine, apex, and leaf attachments. For m spine vertices these choices give the summand of C_N in Equation (6). Each of the $N-2$ generators has three face states, giving at most $3^{N-2}C_N$ descriptions. The $N(N-1)$ descriptions with no retained hinge and head $x_i - x_j$, $i \neq j$, can be combined with the auxiliary functions. By Theorem 7, the second width is the number of retained descriptions plus two; Appendix C gives the complete count.

Formalization (`Lean_17`). The theorem and equation links point to the Lean development at a fixed repository revision. Appendix D prints the network definitions and the main theorem, and describes the verification certificate.

6 Compression and lower bounds

Theorem 7 reduces the second width to the number of retained units in an expansion. This section bounds that number through the dimension of the vocabulary. Let $V_N := \operatorname{span} \mathcal{D}$, and let $A_N \subseteq V_N$ be the span of the auxiliary functions Equation (18). Put

$$q_N := \dim\big(V_N / A_N\big) = \dim V_N - \dim A_N. \tag{67}$$

Theorem 9 (Quotient compression, `Lean_18`, Proof)**.** *For every $N \geq 3$ there are a set $K \subseteq \mathcal{K}_0$ with $|K| \leq q_N$, rational coefficients d_k, and $R \in A_N$ with rational auxiliary coefficients such that $\max_N = \sum_{k\in K} d_k U_k + R$. Hence $\max_N$ is computed by a network with rational weights, zero biases, and widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = q_N + 2. \tag{68}$$

Proof of Theorem 9. Choose a basis of A_N among the auxiliary functions and extend it by face functions to a basis of V_N. The extension adds exactly q_N face functions. For each of them select one normalized description, rather than splitting its coefficient across all descriptions of the same function; this gives K with $|K| \leq q_N$. By Theorem 8, $\max_N \in V_N$, so $\max_N$ is a combination of the selected units and the auxiliary functions. All these functions are continuous and take rational values at rational points, so Lemma 14 gives rational coefficients. Theorem 7 compiles the expansion. □

The retained descriptions together with the auxiliary functions span V_N, so q_N never exceeds the count of Theorem 1. The following lower bound shows that q_N has the same growth.

Theorem 10 (Vocabulary dimension, `Lean_19`, Proof)**.** *For every $N \geq 3$,*

$$q_N \geq (N-2)!. \tag{69}$$

Consequently $\log q_N = \Theta(N \log N)$.

Proof of Theorem 10. Put $b := N - 1$. For every permutation π of $\{1, \dots, N-2\}$, the all-cut path with apex 0, spine $\pi(1), \dots, \pi(N-2), b$, and all edges kept as hinges has the caterpillar function

$$U_\pi(x) = \sigma\Big(x_b - x_0 + \sum_{j=1}^{N-2} \sigma\big(x_{\pi(j)} - x_{\pi(j+1)}\big)\Big), \qquad \pi(N-1) := b. \tag{70}$$

For a permutation ρ, define the *test configuration* $x^\rho(t)$ and the *reference configuration* $y(t)$ by

$$x^\rho_0 = y_0 = 1 + t, \qquad x^\rho_{\rho(j)} = \frac{j}{N-1}, \qquad x^\rho_b = y_b = 1, \qquad y_i = 2 \quad (1 \leq i \leq N-2). \tag{71}$$

Thus the coordinates other than x_0 increase along the path of ρ and end at $x_b = 1$. With $\varepsilon := 1/(2(N-1))$, define the linear functional

$$\Lambda_\rho g := Jg\big(x^\rho\big) - Jg(y), \qquad Jg(z) := g\big(z(\varepsilon)\big) + g\big(z(-\varepsilon)\big) - 2g\big(z(0)\big). \tag{72}$$

The symmetric second difference J vanishes on functions that are affine in t on $[-\varepsilon, \varepsilon]$, and $J\sigma(a+tc) = \varepsilon|c|$ if $a = 0$. For an auxiliary function only $\sigma(\pm(x_0 - x_b))$ has a kink at $t = 0$, in both configurations alike: the other coordinates differ from 1 by at least 2ε. Hence Λ_ρ annihilates A_N. In U_π the inner hinges do not involve x_0, and $x_b - x_0 = -t$ in both configurations. At y the last inner hinge equals $\sigma(2-1) = 1$, so U_π is affine near $t = 0$. At x^ρ the inner sum vanishes exactly when the values increase along the path of π, that is, when $\pi = \rho$; otherwise a descent contributes at least $1/(N-1) = 2\varepsilon$. Therefore

$$\Lambda_\rho U_\pi = \begin{cases} \varepsilon, & \pi = \rho, \\ 0, & \pi \neq \rho. \end{cases} \tag{73}$$

If $\sum_\pi c_\pi U_\pi \in A_N$, applying Λ_ρ gives $c_\rho \varepsilon = 0$ for every ρ. Thus the $(N-2)!$ path units are linearly independent modulo A_N. The upper bound $q_N \leq 3^{N-2} C_N = 2^{O(N \log N)}$ follows from Theorem 1. □

Scope. Theorem 10 bounds the dimension of the vocabulary. It does not bound $s_N := \min\{|J| : \max_N \in \operatorname{span}\{U_j : j \in J\} + A_N\}$, the number of units in the sparsest expansion of $\max_N$, with arbitrary real coefficients. Neither bound concerns unrestricted ReLU networks.

Theorem 11 (Small arities, `Lean_20`, `Lean_21`)**.** *For all real inputs,*

$$\max_3(x) = x_1 + \sigma\big(x_2 - x_1 + \sigma(x_0 - x_2)\big), \tag{74}$$

$$\begin{aligned}\max_4(x) = {} & x_1 - \sigma(x_0 - x_2) - \sigma\big(x_1 + x_2 - x_0 - x_3 + \sigma(x_0 - x_1)\big) \\ & + \sigma\big(x_1 + x_2 - x_0 - x_3 + \sigma(x_0 - x_1) + \sigma(x_0 - x_2)\big) \\ & + \sigma\big(x_3 - x_1 + \sigma(x_0 - x_2) + \sigma(x_2 - x_3)\big).\end{aligned} \tag{75}$$

Hence the network of Theorem 7 computes $\max_3$ *with widths* $(7, 3)$ *and* $\max_4$ *with widths* $(11, 5)$*, with rational weights and zero biases.*

Each outer unit in Equations (74) and (75) is a normalized face function. Lean proves both identities by exhaustive case analysis of the activation pattern on all of $\mathbb{R}^N$. For these arities, smaller networks are known: nesting $\max(s, t) = t + \sigma(s - t)$ gives widths $(6, 3)$ for $\max_4$. The identities illustrate how sparse the expansion in the vocabulary can be.

Exact search in small arity. The script `scripts/vocabulary_search.py` enumerates $\mathcal{K}_0$ and certifies ranks and identities exactly. On each open cell of the order of $x_0, \dots, x_{N-1}$ and 0, every inner hinge and auxiliary function is linear, so every unit has the form $\sigma \circ a$ on a simplicial cone. A combination of such functions vanishes on the cone exactly when all kink coefficients of hyperplanes meeting its interior and the remaining linear part vanish. These finitely many integer conditions certify relations and exclude supports over $\mathbb{Q}$, and hence over $\mathbb{R}$. Evaluations at random integer points modulo a prime give certified lower bounds on ranks. Table 1 summarizes the results. They are computational certificates, not Lean proofs, except for the two networks of Theorem 11.

N	3	4	5	6
h_2 of Theorem 1	5	314	9702	315902
distinct normalized descriptions $\lvert\mathcal{K}_0\rvert$	9	138	1830	26780
$\lvert\mathcal{K}_0\rvert - N(N-1) + 2$	5	128	1812	26752
q_N	1	65	1136	
s_N	1	3	≤ 164	

Table 1: Small-arity data. The counts $\lvert\mathcal{K}_0\rvert$ are exact enumerations, and distinct descriptions give distinct functions. The values q_3, q_4, q_5, s_3, s_4 are exact: lower bounds by rank certificates and upper bounds by relations verified on every cell; $s_4 \geq 3$ excludes every support of at most two units. The bound $s_5 \leq 164$ is an exact expansion found by a mixed-integer search; no matching lower bound is known.

7 Sums of maxima with shared parts

By the Wang–Sun theorem, $f = \sum_{t=1}^{T} \varepsilon_t \max_{d+1} \circ A_t$. Placing T copies of the Caterpillar Net in parallel gives a network for f with widths $T\,h_1$ and $T\,h_2$. Parts of these copies can be shared. The remainder neurons $\sigma(\pm R)$ of the copies merge into a single pair. When the coordinates of the maps A_t are drawn from a fixed finite set of affine functions, the first hidden layer is also shared, and so are the retained second hidden layer neurons of all terms with the same arguments. Such a finite set always exists:

Theorem (Koutschan–Moser–Ponomarchuk–Schicho, [KMPS23, Theorem 2.4])**.** *Every CPWL function* $f : \mathbb{R}^d \to \mathbb{R}$ *is an integral linear combination of maxima of at most* $d + 1$ *affine functions. These affine functions can be chosen among the* constituents *of* f*, the affine functions that coincide with* f *on some nonempty open set.*

Throughout, $N \geq 3$ and h_1, h_2 are the widths Equation (5). The coefficients ε_t are arbitrary real numbers.

Theorem 12 (Shared Caterpillar Nets, `Lean_22`, `Lean_23`, Proof)**.** *Let $\varepsilon_1, \dots, \varepsilon_T \in \mathbb{R}$.*

(a) For arbitrary affine maps $A_1, \dots, A_T : \mathbb{R}^d \to \mathbb{R}^N$, the function $\sum_{t=1}^{T} \varepsilon_t \max_N \circ A_t$ is computed by a two-hidden-layer ReLU network with widths

$$T\,h_1 \qquad \text{and} \qquad T\,(h_2 - 2) + 2. \tag{76}$$

(b) Let $g_1, \dots, g_p : \mathbb{R}^d \to \mathbb{R}$ be affine and let $\alpha_t : \{0, \dots, N-1\} \to \{1, \dots, p\}$ for $1 \leq t \leq T$. Put $S_t := \alpha_t(\{0, \dots, N-1\})$ and $s := |\{S_1, \dots, S_T\}|$. Then

$$f(x) = \sum_{t=1}^{T} \varepsilon_t \max_N\big(g_{\alpha_t(0)}(x), \dots, g_{\alpha_t(N-1)}(x)\big) \tag{77}$$

is computed by a two-hidden-layer ReLU network with widths

$$\binom{p}{2} + p + 1 \qquad \text{and} \qquad s\,(h_2 - 2) + 2. \tag{78}$$

Here $s \leq \min\big(T, \sum_{m=1}^{N} \binom{p}{m}\big)$, and $s \leq \min\big(T, \binom{p}{N}\big)$ if every α_t is injective.

For $T \geq 2$, (a) improves the second width of the parallel network by $2(T-1)$, and (b) by at least $(T-s)(h_2-2) + 2(T-1)$. The first width in (b) does not depend on T. Combined with the representation theorem above, which is not formalized, p can be taken to be at most the number of constituents of f, with $N = d + 1$; maxima of fewer than $d + 1$ constituents repeat an argument, which is allowed in Equation (77). If p is large compared with T, the first hidden layer of (a) is smaller.

Proof of Theorem 12. Step 12.1 (Write the maximum in pair features). By Equations (29) and (120),

$$\max_N(y) = \sum_{k \in K} d_k U_k(y) + \sum_i p_i y_i + \sum_{i<j} s_{ij}\sigma(y_i - y_j), \qquad \sum_i p_i = 1, \tag{79}$$

with $|K| \leq h_2 - 2$ by Equation (121). Each $U_k = U_{h,M}$ applies σ to the coordinate difference $h(y)$ plus hinges $\sigma(y_i - y_j)$. Since $\sum_i p_i y_i = y_0 + \sum_i p_i(y_i - y_0)$, and $y_i - y_j = \sigma(y_i - y_j) - \sigma(y_j - y_i)$ by Equation (8),

$$\max_N(y) = y_0 + \sum_{k \in K} d_k\,\sigma\big(\lambda_k(y)\big) + \lambda_R(y), \tag{80}$$

where λ_k and λ_R are fixed linear combinations of the $N(N-1)$ functions $y \mapsto \sigma(y_i - y_j)$, $i \neq j$.

Step 12.2 (Merge the remainders). Let $y^{(1)}, \dots, y^{(\tau)} : \mathbb{R}^d \to \mathbb{R}^N$, $e_1, \dots, e_\tau \in \mathbb{R}$, and let a first hidden layer $\Psi(x) = \sigma(W_1 x + b_1)$ be such that every coordinate $y_i^{(u)}(x)$ and every hinge $\sigma\big(y_i^{(u)}(x) - y_j^{(u)}(x)\big)$ is a fixed linear combination of the entries of $\Psi(x)$. Substituting $y = y^{(u)}(x)$ into Equation (80) gives row vectors $w_{u,k}$ and w_R with

$$w_{u,k}^{\top}\Psi(x) = \lambda_k\big(y^{(u)}(x)\big), \qquad w_R^{\top}\Psi(x) = \sum_{u=1}^{\tau} e_u\Big(y_0^{(u)}(x) + \lambda_R\big(y^{(u)}(x)\big)\Big). \tag{81}$$

Take second hidden layer rows $w_{u,k}^\top$ ($1 \le u \le \tau$, $k \in K$), $w_R^\top$, and $-w_R^\top$, output weights $e_u d_k$, 1, and -1, and $b_2 = 0$, $b_3 = 0$. By Equation (8), $\sigma(w_R^\top \Psi) - \sigma(-w_R^\top \Psi) = w_R^\top \Psi$, so the network computes

$$\sum_{u=1}^{\tau} e_u \Big(y_0^{(u)} + \sum_{k \in K} d_k \sigma\big(\lambda_k(y^{(u)})\big) + \lambda_R(y^{(u)}) \Big) = \sum_{u=1}^{\tau} e_u \max_N \circ y^{(u)}. \tag{82}$$

Its second hidden layer width is $\tau|K| + 2 \le \tau(h_2 - 2) + 2$. The remainder $\sum_u e_u (y_0^{(u)} + \lambda_R \circ y^{(u)})$ is affine in x plus hinges, and it is represented with only two neurons because it is a linear combination of first hidden layer outputs. The constant terms of affine maps enter only through b_1.

Step 12.3 (Arbitrary affine maps). For (a), take $\tau = T$, $y^{(t)} = A_t$, and $e_t = \varepsilon_t$. Let $\Psi(x)$ stack the vectors $\Phi(A_1 x), \dots, \Phi(A_T x)$, with Φ from Equation (23). Then W_1 stacks the matrices of Φ composed with the linear parts of A_t, and b_1 stacks the images of the translation parts. By Equations (8) and (26), each block recovers all coordinates and all oriented hinges of its term. This gives the first width $T\, h_1$.

Step 12.4 (Shared affine pieces). For (b), write $g(x) := (g_1(x), \dots, g_p(x)) \in \mathbb{R}^p$ and let $\Psi = \Phi \circ g$, where Φ is Equation (23) for p coordinates:

$$\Psi(x) = \big(\sigma(g_1),\ \sigma(-g_1),\ (\sigma(g_a - g_1))_{a>1},\ (\sigma(g_a - g_b))_{a<b}\big)(x). \tag{83}$$

It has $\binom{p}{2} + p + 1$ entries and recovers every g_a and every $\sigma(g_a - g_b)$, $a \ne b$, by Equations (8) and (26). The maximum of a list depends only on the set of its entries, so the summand of Equation (77) for t is $\max_{a \in S_t} g_a$. For each of the s distinct sets S fix one t_S with $S_{t_S} = S$, and put

$$y^{(S)} := \big(g_{\alpha_{t_S}(0)}, \dots, g_{\alpha_{t_S}(N-1)}\big), \qquad e_S := \sum_{t:\, S_t = S} \varepsilon_t, \qquad f = \sum_S e_S \max_N \circ y^{(S)}. \tag{84}$$

Step 12.2 with $\tau = s$ gives Equation (78). The sets S_t are nonempty subsets of $\{1, \dots, p\}$ with at most N elements, and with exactly N elements if α_t is injective. Zero padding realizes the stated widths exactly. □

Further merging. Theorem 12(b) counts at most one neuron for every pair of a set S and a retained unit. Units of different sets may coincide. If $P_\alpha : \mathbb{R}^N \to \mathbb{R}^p$ maps $e_i \mapsto e_{\alpha(i)}$, then $U_{h,M}(g_{\alpha(0)}, \dots, g_{\alpha(N-1)}) = h_{P_\alpha Q_{h,M}}(g(x))$, and $P_\alpha Q_{h,M}$ is the pyramid over the translated zonotope $P_\alpha h + \sum_{\ell_{ij} \in M} [0, e_{\alpha(i)} - e_{\alpha(j)}]$. Units with equal pushed-forward descriptions are equal functions of x; they share one neuron, and their output coefficients add. We did not bound this count in closed form, and it is not formalized.

Theorem 13 (CPWL functions, `Lean_24`, `Lean_25`)**.** *Let $d \ge 2$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be CPWL. Then f has a Wang–Sun representation with some number T of terms. Let p be the number of distinct affine functions occurring in it, and $s \le T$ the number of distinct argument sets of its terms. Then f is computed by two-hidden-layer ReLU networks with widths $\big(T\, h_1,\ T(h_2 - 2) + 2\big)$ and $\big(\binom{p}{2} + p + 1,\ s(h_2 - 2) + 2\big)$, where (h_1, h_2) are the widths of Theorem 1 for $N = d + 1$.*

Here T, p, and s are those of the representation produced by the formal proof of the Wang–Sun theorem; they are not bounded in terms of the pieces of f. With the representation of Koutschan–Moser–Ponomarchuk–Schicho, p is at most the number of constituents of f; this refinement is not formalized. The networks of this section are not claimed to have rational weights; their first hidden layer has the biases of the affine maps.

Formalization (`Lean_26`). The general statement assumes an arbitrary expansion Equation (80) and an arbitrary first hidden layer that spans the required coordinates and hinges. It gives second width $\tau|K| + 2$. The Caterpillar Net supplies such an expansion with $|K| + 2 \le h_2$. Part (b) is formalized with the bound s, and with $\min\big(T, \binom{p}{N}\big)$ for injective α_t; the bound $s \le \sum_{m \le N} \binom{p}{m}$ is not formalized. The Wang–Sun theorem is formalized (Theorem 13); the representation theorem of Koutschan–Moser–Ponomarchuk–Schicho is not.

8 Conclusion

Within its vocabulary, the construction is close to optimal in one sense: the dimension q_N that controls Theorem 9 grows as $2^{\Theta(N \log N)}$. This does not bound the sparsest expansion s_N, which equals 1 and 3 for $N = 3, 4$, nor general networks. The first hidden layer width is asymptotically sharp. Building on the extremal-graph approach of Safran [Saf26], our separate Lean development extends the bound $h_1 \geq \binom{N}{2} - \lfloor 4N/3 \rfloor$ of Lu [Lu26, Theorem 2.13], for $N \geq 4$, to networks with arbitrary affine skip connections. When every first hidden layer weight vector is proportional to some $e_i - e_j$, we sharpen it to $h_1 \geq \binom{N}{2} - N$. We obtain no growing lower bound for h_2 or for s_N; finding smaller exact maximum representations and establishing such lower bounds remain open.

A Canonical coefficients

An exact coefficient rule (`Lean_27`) Let

$$\mathcal{B} := \{S \subseteq \mathcal{D} : S \text{ is a basis of } \operatorname{span} \mathcal{D}\}. \tag{85}$$

This set is nonempty because the finite spanning set $\mathcal{D}$ contains a basis. For $S \in \mathcal{B}$, form

$$G_S := \big(\langle g, h\rangle\big)_{g,h\in S}, \tag{86}$$

$$b_S := \big(\langle g, \max_N\rangle\big)_{g\in S}. \tag{87}$$

using the inner product

$$\langle f, g\rangle := \int_{[-1,1]^N} f(x)g(x)\,dx. \tag{88}$$

If a linear combination of basis functions has zero squared integral, continuity makes it zero on the cube and positive homogeneity makes it zero on $\mathbb{R}^N$. Linear independence forces its coefficients to vanish; hence G_S is positive definite and invertible. For each $g \in \mathcal{D}$, average over the bases containing g:

$$c_g := \frac{1}{|\mathcal{B}|} \sum_{\substack{S\in\mathcal{B}\\ g\in S}} (G_S^{-1} b_S)_g. \tag{89}$$

Proof of Lemma 6. Fix $S \in \mathcal{B}$. By Theorem 8, there is $d \in \mathbb{R}^S$ such that

$$\max_N = \sum_{g\in S} d_g g. \tag{90}$$

Taking the inner product with each $h \in S$ gives

$$\begin{aligned} (b_S)_h &= \sum_{g\in S} \langle h, g\rangle d_g = (G_S d)_h, \\ d &= G_S^{-1} b_S. \end{aligned} \tag{91}$$

Interchanging the finite sums in Equation (89) therefore yields

$$\sum_{g\in\mathcal{D}} c_g g = \frac{1}{|\mathcal{B}|} \sum_{S\in\mathcal{B}} \sum_{g\in S} (G_S^{-1} b_S)_g\, g = \frac{1}{|\mathcal{B}|} \sum_{S\in\mathcal{B}} \max_N = \max_N . \tag{92}$$

□

Averaging over all bases avoids an arbitrary choice of basis and gives a finite, nonrecursive coefficient prescription. We do not claim an efficient algorithm for evaluating these coefficients or a bound on their bit complexity.

Rational coefficients without integration. The inner product is used only to name one coefficient vector. Rationality follows from evaluations alone.

Lemma 14 (Rational coefficients, `Lean_28`, Proof)**.** *Let $g_1, \dots, g_m : \mathbb{R}^N \to \mathbb{R}$ be continuous and linearly independent, and assume that every g_i takes rational values at points of $\mathbb{Q}^N$. If $f = \sum_i d_i g_i$ also takes rational values on $\mathbb{Q}^N$, then every d_i is rational. Consequently, every function in the span of a finite family of such functions, rational on $\mathbb{Q}^N$, is a rational combination of that family.*

Proof of Lemma 14. A combination of the g_i vanishing on the dense set $\mathbb{Q}^N$ vanishes everywhere by continuity, so the g_i are independent on $\mathbb{Q}^N$. The rational rows $(g_i(y))_i$, $y \in \mathbb{Q}^N$, therefore span $\mathbb{Q}^m$: otherwise a nonzero rational vector c would satisfy $\sum_i c_i g_i(y) = 0$ for all $y \in \mathbb{Q}^N$. Hence there are finitely many points $y_1, \dots, y_r$ and rational numbers $a_1, \dots, a_r$ with $\sum_k a_k g_i(y_k) = \delta_{ij}$ for all i, and

$$d_j = \sum_i d_i \sum_k a_k g_i(y_k) = \sum_k a_k f(y_k) \in \mathbb{Q}. \tag{93}$$

For the second statement, choose a basis of the span inside the family and apply the first statement to it. □

The dictionary functions are continuous and rational on $\mathbb{Q}^N$: they are compositions of σ with integer combinations of coordinates, and so is $\max_N$. Two coefficient rules follow.

Version 1: the fixed rule (`Lean_29`). The rule Equation (89) assigns one coefficient vector to the linearly dependent family $\mathcal{D}$ without choices. Its coefficients are rational: on each basis $S \in \mathcal{B}$, the vector $G_S^{-1} b_S$ is the unique coordinate vector of $\max_N$ by Equation (91), which is rational by Lemma 14, and c_g averages these.

Version 2: one basis (`Lean_30`). Select a single basis $S \subseteq \mathcal{D}$ of $\operatorname{span} \mathcal{D}$ and take the coordinates of $\max_N$ in S. They are rational by Lemma 14, and no integral is needed. The rule depends on the choice of S; uniqueness holds only for coordinates in a fixed basis, not for representations in all of $\mathcal{D}$.

The network of Theorem 1 uses Version 1; hence its weights are rational.

For network construction, use each distinct pair (h', M) from Equation (21) once, together with each auxiliary function in Equation (18). Split c_g equally among all pairs and auxiliary functions representing g. Their total contribution remains $c_g g$. Normalization then merges equal pairs by adding their coefficients.

B Supporting proofs for the dictionary theorem

B.1 Residual reduction

We prove the inner induction used in the simplex-face argument of Section 4. Fix a nonempty vertex set T with $|T| \geq 2$, and assume that every proper simplex face on a subset of T is already in the dictionary span, as in the outer induction there.

Proof of Claim 8.2. We argue by induction on j.

Start ($j = 0$). By definition, $\varepsilon_T^{(0)} = \varepsilon_T$. Its dimension bound and support-function expression were established above, and the removed part is zero.

Step ($j < |T| - 2$). Assume the directions for the first j stages have been chosen and the claim holds at stage j. Put $k := |T| - 2 - j \geq 1$. We require each new direction u to be nonorthogonal to the simplex direction space $\operatorname{dir} \Delta_T := \operatorname{span}\{v - w : v, w \in T\}$. Its values on T are then not all equal, so the maximizing vertices form a nonempty proper subset:

$$\begin{aligned} A_u &:= \{v \in T : \langle u, v\rangle = h_{\Delta_T}(u)\}, \qquad \varnothing \neq A_u \subsetneq T, \\ \partial_u h_{\Delta_T} &= h_{\Delta_{A_u}} \in \operatorname{span} \mathcal{D}. \end{aligned} \tag{94}$$

The last membership uses the outer induction hypothesis, since $|A_u| < |T|$. An orthogonal direction would leave h_{Δ_T} unchanged and would not allow this use of the hypothesis.

Choose an auxiliary expression for $\varepsilon_T^{(j)}$ by support functions of polytopes of dimension at most k, and collect their distinct k-dimensional direction spaces. For each such space V, choose $u_V \in V^\perp$ not orthogonal to $\operatorname{dir} \Delta_T$ or to any other collected space V'. This is possible because $\dim \Delta_T = |T| - 1 > k$: each of $\operatorname{dir} \Delta_T$ and V' contains a vector outside V, whose nonzero projection onto $V^\perp$ is not orthogonal to that vector. Adding sufficiently small multiples of these projections avoids each orthogonality condition in turn while preserving those already avoided. Choose $E_{T,j} := \{u_V : V \text{ is a collected direction space}\}$, with $E_{T,j} = \varnothing$ if there are none. In the sum of derivatives over $E_{T,j}$, each term coming from a k-dimensional polytope is preserved exactly once. Every other derivative of that term is the support function of a proper face, while terms of lower dimension remain in $\mathcal{L}_{k-1}$. Subtracting this derivative sum therefore cancels all k-dimensional terms. By the recursive definition Equation (47), the resulting error satisfies $\varepsilon_T^{(j+1)} \in \mathcal{L}_{k-1} = \mathcal{L}_{|T|-3-j}$.

Now compute these same derivatives using the support-function expression in the claim. The derivative of h_{Δ_T} lies in the dictionary span by Equation (94). Differentiating the support function of an already proper simplex face still gives a support function of a proper face of Δ_T, so the same induction hypothesis applies. Differentiating the support function of a face of $Q_{\pi,S}$ gives a support function of a face of the same polytope, whose support functions lie in the dictionary span, and linear functions remain linear. Thus the derivative sum lies in the dictionary span. By Equation (47), $\varepsilon_T - \varepsilon_T^{(j+1)}$ is the sum of the previously removed part and this derivative sum, so it also lies in the dictionary span. Subtracting the derivative sum preserves the stated types of support-function terms. This completes the proof step.

□

B.2 Reconstruction from opposite exposed faces

We justify Step 8.2: for $N \geq 4$, the bounds Equation (49) imply $f \in \mathcal{L}_{N-2}$. We first subdivide the caterpillar polytopes, then prove Claim 8.3, and finally apply it to the resulting weighted simplex sum.

The base of each $Q_{\pi,S}$ is an affine cube. Order its edge vectors by a bijection $\omega : \{1, \dots, N-2\} \to L_{\pi,S}$, add them one at a time, and adjoin the apex:

$$K_{\pi,S,\omega} := \operatorname{conv}\left(\{0\} \cup \left\{h_{\pi,S} + \sum_{a=1}^{k} \omega(a) : 0 \leq k \leq N-2\right\}\right). \tag{95}$$

These $(N-2)!$ simplices are obtained by coning the *Freudenthal–Kuhn triangulation* of the base to the apex [Fre42, Kuh60]: in cube coordinates, each base simplex occupies one ordering region $1 \geq t_{\omega(1)} \geq \cdots \geq t_{\omega(N-2)} \geq 0$. For a hyperplane cut $P = P_+ \cup P_-$, the support-function identity

$$h_P + h_{P_+ \cap P_-} = h_{P_+} + h_{P_-} \tag{96}$$

shows that subdivision changes the sum of support functions only by support functions of the cutting sections. Hence

$$h_{Q_{\pi,S}} - \sum_{\omega} h_{K_{\pi,S,\omega}} \in \mathcal{L}_{N-2}. \tag{97}$$

Substituting Equation (97) into Equation (39) expresses f as a finite linear combination of support functions of $(N-1)$-dimensional simplices in H_0, plus a remainder in $\mathcal{L}_{N-2}$. The simplex Δ contributes its support function with coefficient -1, and the linear apex terms are included in the remainder.

For a simplex P, an outward facet normal ν_F exposes the facet F on one side and its opposite vertex v_F on the other. Thus

$$D_{\nu_F} h_P(x) = h_F(x) - \langle v_F, x \rangle. \tag{98}$$

Directions exposing no facet on either side give support functions of faces of dimension at most $N-3$. Consequently, Equation (49) says that each weighted, signed sum of parallel-facet support functions belongs to $\mathcal{L}_{N-3}$.

Our assumption controls the weighted facet contributions, whereas our conclusion concerns f. To recover the original sum modulo $\mathcal{L}_{N-2}$, we use the same linear operator for every simplex, so cancellations between facet contributions survive reconstruction. Use the operator $\mathcal{T}$ defined in Equation (50). The sign in this definition concerns the input x. To interpret it geometrically, first suppose that every point $p \in P$ satisfies $p_1 \geq 0$, and write $\mathrm{pr}(p) := p - p_1 e_1$ for projection onto $p_1 = 0$. Since $\langle p, x \rangle - \langle \mathrm{pr}(p), x \rangle = p_1 x_1$, each point dominates its projection when $x_1 \geq 0$, and its projection dominates when $x_1 < 0$. Consequently,

$$\mathcal{T} h_P(x) = \max\{h_P(x), h_{\mathrm{pr}\, P}(x)\} = h_{\mathrm{conv}(P \cup \mathrm{pr}\, P)}(x). \tag{99}$$

This convex hull fills the segments with endpoints p and its projection, for $p \in P$, so its dimension is at most $\dim P + 1$. For an arbitrary P, choose $c \geq 0$ such that $P + ce_1$ lies in $p_1 \geq 0$. Translation adds cx_1 to the support function, and $\mathcal{T}(x_1) = \sigma(x_1)$, giving

$$\mathcal{T} h_P(x) = h_{\mathrm{conv}((P+ce_1) \cup \mathrm{pr}(P+ce_1))}(x) - c\sigma(x_1). \tag{100}$$

The correction $\sigma(x_1)$ is the support function of the segment $[0, e_1]$. Thus linearity gives

$$\mathcal{T} \mathcal{L}_k \subseteq \mathcal{L}_{k+1}. \tag{101}$$

Proof of Claim 8.3. Let $k = |\{F \text{ facet of } P : \langle \nu_F, e_1 \rangle \neq 0\}|$ count facets not parallel to e_1. Boundedness ensures both nonzero signs occur, so $k \geq 2$.

Start ($k = 2$). There is one upper facet F^+ and one lower facet F^-. Every line parallel to e_1 that meets P has its two intersection endpoints on these facets. Thus $h_P(x) = h_{F^+}(x)$ when $x_1 \geq 0$, and $h_P(x) = h_{F^-}(x)$ when $x_1 < 0$. When $x_1 \geq 0$, $\mathcal{T}$ evaluates at x, so subtracting $\mathcal{T}(h_{F^+} - h_{F^-})$ from h_P leaves h_{F^-}. When $x_1 < 0$, $\mathcal{T}$ evaluates both facet support functions at $x - x_1 e_1$, where they agree because the facets have the same projection. Their difference is therefore zero, again leaving h_{F^-}. The reconstruction error is therefore $h_{F^-} \in \mathcal{L}_{N-2}$, proving the claim for $k = 2$.

Step ($k > 2$). Assume the claim for $k-1$, and let P have k facets not parallel to e_1. Two of them have the same nonzero sign of $\langle \nu_F, e_1 \rangle$. Their intersection is an $(N-3)$-dimensional face C. Let $W := \mathrm{span}\{p - q : p, q \in C\}$, and project orthogonally within H_0 onto $W^\perp$. This is a two-dimensional plane, since $\dim H_0 = N-1$ and $\dim W = N-3$. All points of C have the same projection c, because their differences lie in W. There are just two vertices of P outside C, so the projected simplex is a triangle with vertices c and the projections of those two vertices. The chosen facets project to its two sides meeting at c.

The chosen facet normals are orthogonal to W, so their signs against e_1 equal their signs against its projection. This projected direction is nonzero, since the chosen facets are not parallel to e_1. If both sides are upper, moving from c opposite to this direction enters the triangle; if both are lower, moving in this direction

enters it. Continuing meets the interior of the opposite edge. The inverse image of the line through c in this direction is a hyperplane in H_0 containing C and parallel to e_1. It cuts the edge whose endpoints are the two vertices outside C at an interior point and divides P into two $(N-1)$-dimensional simplices. Each keeps one of the two chosen facets and replaces the other by the new facet parallel to e_1. Its other facets retain their normal directions, so each new simplex has exactly $k-1$ facets not parallel to e_1.

Write $E(Q)$ for the reconstruction error on the left of Equation (51). Let P_+ and P_- be the two new simplices, let $R := P_+ \cap P_-$ be their common cutting facet, and write F_1, F_2 for the chosen facets and $R_F := R \cap F$ for every other facet of P. With $s_F := \operatorname{sgn}\langle \nu_F, e_1 \rangle$, applying Equation (96) to P and to every split facet gives the exact residual identity

$$E(P) = E(P_+) + E(P_-) - h_R + \mathcal{T}\left(\sum_{\substack{F \text{ facet of } P \\ F \notin \{F_1, F_2\}}} s_F h_{R_F} \right). \tag{102}$$

Indeed, the inherited facets retain their signs, while the two copies of R are parallel to e_1 and therefore have sign zero. By the induction hypothesis, $E(P_+), E(P_-) \in \mathcal{L}_{N-2}$; moreover, $\dim R = N-2$ and $\dim R_F \le N-3$. Equation Equation (101) therefore puts every term on the right of Equation (102) in $\mathcal{L}_{N-2}$. This proves the claim for k and completes the induction.

□

Conclusion for f. Apply Equation (51) to every simplex support-function term in the expansion of f, keeping its original coefficient. A facet F parallel to e_1 has $\langle \nu_F, e_1 \rangle = 0$, so its term in the reconstruction formula is zero and can be omitted. For each remaining occurrence of a facet, choose ν_F outward from its simplex and put

$$u_F := \frac{\nu_F}{\langle \nu_F, e_1 \rangle}, \qquad \langle u_F, e_1 \rangle = 1. \tag{103}$$

Thus the coordinate with index 1 is normalized to 1; the coordinate with index 0 remains zero because $u_F \in H_0$. Oppositely oriented parallel normals give the same u_F. Only the normal is rescaled: the support function and its coefficient are unchanged.

Let $\mathcal{U}$ be the finite set of vectors u_F obtained this way. For each $u \in \mathcal{U}$, define G_u by adding all facet support-function terms with $u_F = u$, retaining their signs and weights. An occurrence of F as a facet of $K_{\pi,S,\omega}$ contributes $(-1)^{N+|S|} \operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F/N!$, while an occurrence as a facet of Δ contributes $-\operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F$. These are precisely the terms with the same normalized normal, including all repetitions.

More explicitly, define ρ to be the sum of the linear apex terms and the weighted subdivision errors in Equation (97). Then

$$f = \frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,N-2\}} (-1)^{N+|S|} \sum_{\omega} h_{K_{\pi,S,\omega}} - h_\Delta + \rho, \qquad \rho \in \mathcal{L}_{N-2}. \tag{104}$$

For a simplex P, collect its facets with normalized normal u as

$$B_u(P) := \sum_{\substack{F \text{ facet of } P \\ \langle \nu_F, e_1 \rangle \neq 0,\ u_F = u}} \operatorname{sgn}\langle \nu_F, e_1 \rangle h_F. \tag{105}$$

An empty sum is zero. If $\langle \nu_F, e_1 \rangle > 0$, the direction u_F exposes F; if it is negative, $-u_F$ exposes F. The opposite face is a single vertex, so Equation (98) gives the signed facet term up to a linear function. If neither u nor $-u$ exposes a facet of P, both exposed faces have dimension at most $N-3$. Thus

$D_u h_P - B_u(P) \in \mathcal{L}_{N-3}$. Applying this to the full weighted expansion, with the same permutations and nonempty cut sets as in Equation (39), gives

$$\begin{aligned} D_u f - G_u = \sum_{\pi, S \neq \varnothing} \frac{(-1)^{N+|S|}}{N!} \sum_{\omega} \left(D_u h_{K_{\pi,S,\omega}} - B_u(K_{\pi,S,\omega}) \right) \\ - \left(D_u h_\Delta - B_u(\Delta) \right) + D_u \rho \in \mathcal{L}_{N-3} \qquad (u \in \mathcal{U}). \end{aligned} \tag{106}$$

Together with Equation (49), relation Equation (106) puts every G_u in $\mathcal{L}_{N-3}$. By linearity, the sum of the reconstructed facet terms is $\mathcal{T}(\sum_{u \in \mathcal{U}} G_u)$, which lies in $\mathcal{L}_{N-2}$ by Equation (101). Its difference from f consists of the reconstruction and subdivision remainders, also in $\mathcal{L}_{N-2}$. Hence $f \in \mathcal{L}_{N-2}$. This proves the implication used in Step 8.2.

B.3 Coordinate deletion

Proof of Claim 8.5. We now determine which terms of the signed sum must be retained for $u = (0, 1, \ldots, 1)$. Only label 0 has a different coordinate value from the others. Consequently, a tree-edge generator is nonorthogonal to u exactly when one of its endpoint labels is 0.

If label 0 is the apex label, all tree-edge generators are orthogonal to u. The maximizing face is the whole base $B_{\pi,S}$ and the minimizing face is $\{0\}$, so $D_u h_{Q_{\pi,S}} = h_{B_{\pi,S}} \in \mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. If label 0 is a tree vertex of degree at least two, at least two segment generators are nonorthogonal to u. The preceding dimension bound then gives $D_u h_{Q_{\pi,S}} \in \mathcal{L}_{N-3}$ as well. We can therefore discard both kinds of term: it suffices to retain descriptions in which label 0 is a tree vertex of degree one.

To express this condition in the permutation notation, let $p := \pi^{-1}(0)$ be the index of label 0 in the full order $(\pi(0), \ldots, \pi(N-1))$. When $p \geq 1$, its tree position is $p - 1$. A degree-one vertex is either an endpoint of the spine with no attached leaves or a leaf outside the spine. The retained descriptions are thus exactly

$$\begin{aligned} &p = 1, \quad 1 \in S && \text{(root endpoint)}, \\ &2 \leq p \leq N-1, \quad p-1 \notin S && \text{(leaf outside the spine)}, \\ &p = N-1, \quad N-2 \in S && \text{(last spine endpoint)}. \end{aligned} \tag{107}$$

Indeed, the root has only one child precisely when $1 \in S$; a nonroot spine vertex has a parent and also a child unless it occupies the final tree position. Deleting label 0 from the full order now gives an order $\bar{\pi}$ mapping positions $\{0, \ldots, N-2\}$ bijectively to labels $\{1, \ldots, N-1\}$.

For $0 \leq i \leq N-3$, let ι_i insert a gap at the prospective pendant position $i+1$. Given a lower cut set $T \subseteq \{1, \ldots, N-3\}$, the three active insertion rules are

$$\begin{aligned} \iota_i(t) &:= \begin{cases} t, & t < i+1, \\ t+1, & t \geq i+1, \end{cases} \\ \text{first end:} \quad & (p, S) = \left(1, \{1\} \cup \{t+1 : t \in T\}\right), \\ \text{pendant } i\text{:} \quad & (p, S) = \left(i+2, \{\iota_i(t) : t \in T\}\right) \qquad (0 \leq i \leq N-3), \\ \text{last end:} \quad & (p, S) = \left(N-1, T \cup \{N-2\}\right). \end{aligned} \tag{108}$$

In every row the upper permutation is uniquely obtained by inserting label 0 at position p in $\bar{\pi}$. Conversely, with t restricted to $\{1, \ldots, N-3\}$, the deletion rules are $T = \{t : t+1 \in S\}$ for the first end, $T = \{t : \iota_i(t) \in S\}$ for pendant i, and $T = S \setminus \{N-2\}$ for the last end. These operations are inverse. The three cases in Equation (107) correspond respectively to the first-end row, the pendant row with $i = p-2$, and the last-end row. Thus a nonempty T has $N-2$ pendant and two endpoint preimages, whereas $T = \varnothing$ has only the two endpoints because upper cut sets are nonempty.

Extend the lower caterpillar notation to $T = \varnothing$ by setting its last spine position and every parent equal to 0, so its head begins at the root $\bar{\pi}(1)$. For every lower label order $\bar{\pi}$ put

$$\begin{aligned} C_{\bar{\pi},T}(x) &:= x_{\bar{\pi}(0)} + U_{\bar{\pi},T}(x), \\ q_T &:= \max(\{0\} \cup T), \qquad p_T(j) := \max(\{0\} \cup \{t \in T : t < j\}). \end{aligned} \tag{109}$$

For each upper preimage (π, S) of $(\bar{\pi}, T)$, the insertion identities and the literal factorial coefficients are

$$\begin{aligned} \text{pendant:} \quad & \frac{(-1)^{N+|S|}}{N!} = -\frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \quad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) + C_{\bar{\pi},T} \in \mathcal{L}_1, \\ \text{endpoint:} \quad & \frac{(-1)^{N+|S|}}{N!} = \frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \qquad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) - C_{\bar{\pi},T} \in \mathcal{L}_1. \end{aligned} \tag{110}$$

Thus every one of the $N - 2 + 2 = N$ preimages contributes $1/N$ of the lower summand modulo $\mathcal{L}_{N-3}$, and together they recover its coefficient in F_{N-1}.

The extra terms are the two endpoint fibres whose deletion leaves $T = \varnothing$, hence a rooted star. By Equation (110), each contributes $1/N$ of the weighted lower summand modulo $\mathcal{L}_1$. For a fixed lower order $\bar{\pi}$, the nonroot positions are $J = \{0, 2, \dots, N-2\}$, including the original apex position 0. For $a \in J$, let $\bar{\pi}^{(a)}$ exchange positions 0 and a, keeping the root at position 1; the exchange at $a = 0$ is the identity. Put

$$\begin{aligned} q := |J| = N - 2, \qquad & d_a := x_{\bar{\pi}(1)} - x_{\bar{\pi}(a)} \quad (a \in J), \\ T_A := \sigma\left(\sum_{a\in A} d_a + \sum_{a\in J\setminus A} \sigma(d_a)\right) \quad & (A \subseteq J), \\ C_{\bar{\pi}^{(a)},\varnothing} = x_{\bar{\pi}(a)} + T_{\{a\}} \quad & (a \in J). \end{aligned} \tag{111}$$

The Boolean alternating identity gives

$$\sum_{A\subseteq J} (-1)^{|A|} T_A = 0 \quad \Longrightarrow \quad \sum_{a\in J} T_{\{a\}} = T_\varnothing + \sum_{\substack{A\subseteq J \\ |A|\geq 2}} (-1)^{|A|} T_A. \tag{112}$$

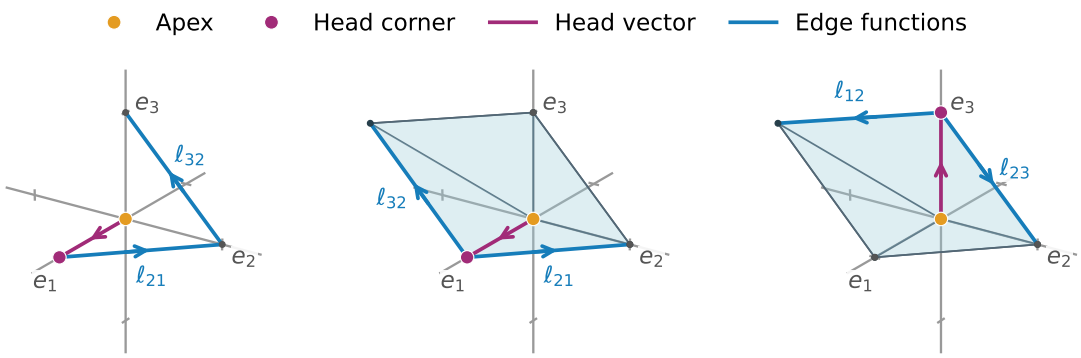
The empty term is a sum of support functions of segments. For $|A| \geq 2$, the polytope with support function T_A has one head direction and $q - |A|$ segment directions, hence dimension at most $1 + q - |A| \leq q - 1 = N - 3$. Thus the singleton sum lies in $\mathcal{L}_{N-3}$; the sum of the linear apex coordinates lies in $\mathcal{L}_0$. Each fixed-position exchange is a bijection on lower label orders, so

$$q \sum_{\bar{\pi}} C_{\bar{\pi},\varnothing} = \sum_{\bar{\pi}} \sum_{a\in J} C_{\bar{\pi}^{(a)},\varnothing} \in \mathcal{L}_{q-1}. \tag{113}$$

Since $q > 0$, the full empty-cut sum belongs to $\mathcal{L}_{q-1} = \mathcal{L}_{N-3}$. The empty-cut coefficient $(-1)^{N-1}/(N-1)!$ and the factor $2/N$ from the two endpoint fibres are scalars, so the extra star sum lies there as well. This proves Claim 8.5. □

C Network construction and width bounds

Proof of Theorem 1. Fix $N \geq 3$. Start with the canonical expansion Equation (20) and distribute its coefficients as described in Appendix A.



(a) Coordinate differences (b) Original description (c) Normalized description

Figure 3: Coordinate differences and the caterpillar polytope for $\pi = (0, 3, 2, 1)$ and $S = \{1, 2\}$. (a) The head $h = e_1 - e_0$ and edge functions $L = \{\ell_{32} = e_3 - e_2, \ell_{21} = e_2 - e_1\}$ drawn between their labelled endpoints. (b) The same generators translated to the base corner h of $Q_{h,L}$. (c) Normalization (Section 5) gives $h^+ = h + \ell_{32} + \ell_{21} = e_3 - e_0$ and $L^+ = \{\ell_{12} = e_1 - e_2, \ell_{23} = e_2 - e_3\}$, leaving the polytope unchanged. Purple arrows represent the head (the apex–last spine connection before normalization); blue arrows represent the spine edge functions. Orange dots mark the polytope apex and purple dots the chosen head corner. The grey coordinate axes meet at the origin, representing e_0 after restriction to $x_0 = 0$; the positive unit points are labelled e_1, e_2, e_3.

Step 1.1 (Count the caterpillar face functions). Use the adjusted heads and normalized descriptions from Section 5. Equal descriptions can share a second hidden layer neuron, with their output coefficients added; counting descriptions therefore gives an upper bound.

For a caterpillar description (π, S), the spine endpoints are labelled $\pi(1)$ and $\pi(q+1)$, and every label other than $\pi(1)$ and the apex is the child of exactly one edge. Hence the normalized description of $U_{\pi,S}$ is

$$\begin{aligned} h^+_{\pi,S}(x) &= \sum_{\ell_{ij} \in L_{\pi,S}} x_{\max(i,j)} + x_{\pi(1)} + x_{\pi(q+1)} - \sum_{i=0}^{N-1} x_i, \\ L^+_{\pi,S} &= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in L_{\pi,S}\}. \end{aligned} \tag{114}$$

This depends only on the undirected tree, apex, and unordered spine endpoints. For m spine vertices, choose the vertex set in $\binom{N}{m}$ ways, its order up to reversal in $m!/2$ ways, the apex in $N - m$ ways, and each of the $N - m - 1$ leaf attachments in m ways. Reversal has no fixed point since $m \geq 2$, so Equation (6) gives

$$\left|\{(h^+_{\pi,S}, L^+_{\pi,S}) : (\pi, S) \text{ a caterpillar description}\}\right| \leq C_N. \tag{115}$$

Reversing an edge swaps its omitted and linear states. Thus choosing states before or after normalization gives the same family $\mathcal{K}_0$ of descriptions, with $|\mathcal{K}_0| \leq 3^{N-2} C_N$. Write $U_k := U_{h,M}$ for $k = (h, M)$. We merge equal descriptions; distinct descriptions may still represent the same function.

Among them are the descriptions $(\ell_{ij}, \varnothing)$ representing $x \mapsto \sigma(x_i - x_j)$, with $i \neq j$: for the all-cut path $S = \{1, \ldots, r\}$ with apex $\pi(0) = j$ and last spine vertex $\pi(r+1) = i$, omitting every edge function gives

$$U^0_{\pi,S} = \sigma \circ \ell_{ij} \qquad (0 \leq i \neq j < N). \tag{116}$$

Let $\mathcal{P} \subseteq \mathcal{K}_0$ consist of descriptions with empty hinge set and head a coordinate difference. The distinct heads ℓ_{ij} give $|\mathcal{P}| \geq N(N-1)$; their contributions can be combined with the auxiliary terms. Let d_k be the sum

of coefficients assigned to pairs (h', M) whose normalized description is k. Discard zero coefficients and remove these descriptions:

$$\mathcal{K}_{\text{act}} := \{k \in \mathcal{K}_0 : d_k \neq 0\}, \tag{117}$$

$$K := \mathcal{K}_{\text{act}} \setminus \mathcal{P}. \tag{118}$$

By Equation (8), every unit indexed by $\mathcal{P}$ lies in the span of the auxiliary functions and needs no separate second hidden layer neuron. Since $K \subseteq \mathcal{K}_0 \setminus \mathcal{P}$,

$$|K| \leq 3^{N-2}C_N - N(N-1). \tag{119}$$

Step 1.2 (Compile the network and establish its widths). Let R collect the weighted auxiliary functions and the terms $d_k U_k$ with $k \in \mathcal{K}_{\text{act}} \cap \mathcal{P}$. Then

$$\max_N = \sum_{k \in K} d_k U_k + R. \tag{120}$$

Equation (8) puts R in the auxiliary span. Theorem 7 compiles this expansion; its weights are rational because the coefficients are rational (Lemma 14). By Equation (119), the second hidden layer width is at most

$$|K| + 2 \leq 3^{N-2}C_N - N(N-1) + 2 = h_2. \tag{121}$$

Zero padding realizes the stated widths exactly.

For the growth rates, count the ordered choices of m spine labels:

$$\binom{N}{m}\frac{m!}{2} = \frac{N!}{2(N-m)!} \leq N^m \qquad (2 \leq m \leq N-1). \tag{122}$$

Using $N - m \leq N$ and $m^{N-m-1} \leq N^{N-m-1}$ in each summand of Equation (6) gives

$$\begin{aligned} C_N &\leq \sum_{m=2}^{N-1} N^m \, N \, N^{N-m-1} = (N-2)N^N \\ &\leq N^{N+1} \leq N^{4N}. \end{aligned} \tag{123}$$

Since $3^{N-2} \leq N^N$ for $N \geq 3$,

$$h_2 \leq 3^{N-2}(N-2)N^N + 2 \leq 2N^{5N}, \tag{124}$$

$$h_1 \leq 4N^2. \tag{125}$$

Taking logarithms gives the asymptotic bounds in Theorem 1. □

D Lean definitions and main theorem

The real parameter arrays in Equation (4) have dimensions

$$\begin{aligned} &W_1 \in \mathbb{R}^{h_1 \times N}, \quad b_1 \in \mathbb{R}^{h_1}, \\ &W_2 \in \mathbb{R}^{h_2 \times h_1}, \quad b_2 \in \mathbb{R}^{h_2}, \\ &w_3 \in \mathbb{R}^{h_2}, \qquad b_3 \in \mathbb{R}, \end{aligned} \tag{126}$$

Sources (`Definitions.lean`, `Theorems.lean`). The listings specify the network, widths, and correctness for every real input. They reproduce declarations from these files at revision `9eff36af7b8d`, omitting imports, documentation comments, and the theorem's proof body; these remain in the linked sources.

D.1 Network and width definitions

```
namespace MaxCaterpillar

open BigOperators

abbrev Point (N : ℕ) := Fin N → ℝ

noncomputable def relu (t : ℝ) : ℝ := max 0 t

structure Network (N h₁ h₂ : ℕ) where
  W₁ : Fin h₁ → Fin N → ℝ
  b₁ : Fin h₁ → ℝ
  W₂ : Fin h₂ → Fin h₁ → ℝ
  b₂ : Fin h₂ → ℝ
  w₃ : Fin h₂ → ℝ
  b₃ : ℝ

noncomputable def Network.eval {N h₁ h₂ : ℕ} (net : Network N h₁ h₂)
    (x : Point N) : ℝ :=
  (∑ k, net.w₃ k * relu
    ((∑ j, net.W₂ k j * relu
      ((∑ i, net.W₁ j i * x i) + net.b₁ j)) + net.b₂ k))
    + net.b₃

def Network.HasRationalWeights {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  (∀ j i, ∃ q : ℚ, (q : ℝ) = net.W₁ j i) ∧
    (∀ k j, ∃ q : ℚ, (q : ℝ) = net.W₂ k j) ∧
    ∀ k, ∃ q : ℚ, (q : ℝ) = net.w₃ k

def Network.HasZeroBiases {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  net.b₁ = 0 ∧ net.b₂ = 0 ∧ net.b₃ = 0

noncomputable def Maximum (N : ℕ) [NeZero N] (x : Point N) : ℝ :=
  Finset.univ.sup'
    ⟨⟨0, Nat.pos_of_ne_zero (NeZero.ne N)⟩,
      Finset.mem_univ _⟩ x

def firstWidth (N : ℕ) : ℕ := N.choose 2 + N + 1

def coneCount (N : ℕ) : ℕ :=
  ∑ m in Finset.Icc 2 (N - 1),
```

```
    N.choose m * (m.factorial / 2) * (N - m)
      * m ^ (N - m - 1)

def secondWidth (N : ℕ) : ℕ :=
  coneCount N * 3 ^ (N - 2) - N * (N - 1) + 2

end MaxCaterpillar
```

D.2 The main theorem

The assumption `[NeZero N]` expresses positive arity. For $N \geq 3$ the statement is Theorem 1; the natural-number width definitions also cover $N = 1, 2$.

```
theorem maximum_width_rational (N : ℕ) [NeZero N] :
    ∃ net : Network N (firstWidth N) (secondWidth N),
      (∀ x, net.eval x = Maximum N x) ∧
        net.HasRationalWeights ∧ net.HasZeroBiases
```

D.3 Version 1 and Version 2 declarations

Improvements were added beside the existing development rather than replacing it. In the Lean sources, superseded declarations are kept and commented as Version 1, and the new ones as Version 2.

- *Main theorem.* Version 1, `maximum_width`, asserts a real-parameter network of the stated widths. Version 2, `maximum_width_rational`, adds rational weights and zero biases. For $N \geq 3$ both use the same network `network`.
- *Coefficients.* Version 1 is the inverse-Gram average `coefficient`; `coefficient_rational` proves it rational. Version 2, `rational_coefficients`, uses one basis and the evaluation argument of Lemma 14.
- *Compilation and widths.* Version 1 compiles the canonical coefficients only (`network`, `construction_width`). Version 2 compiles every expansion (`compilation`), and gives the widths of Theorems 9 and 11.

The Version 1 statements remain checked.

The keyword `noncomputable` allows classical definitions of real quantities. The theorem is an exact representation statement; it does not by itself assert an efficient algorithm for extracting numerical weights.

D.4 Verification scope

The repository's verification certificate, dated 22 October 2026, records a build with Lean 4.4.0 and checks 44 public statements against their exact types. Its permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. The checked interface includes dictionary sufficiency, the canonical expansion and its rationality, network correctness with rational weights and zero biases, the width bounds, the compilation and compression theorems, the lower bound $q_N \geq (N-2)!$, the small-arity networks, the Wang–Sun theorem, and the network theorems for sums of maxima and CPWL functions. The exact searches of Table 1 are not part of the certificate. The certificate and its source hashes are available in the linked verification record. The mathematical arguments are provided in the main text and Appendices A to C; the formalization is an additional certificate of the formal statements.

AI methodology statement

Claude Code and Codex were used for mathematical exploration and proof search, development of the Lean formalization, and drafting and revision of the manuscript. Their use materially affected the construction and proofs in Sections 3 to 7 and Appendices A to D. The author set the research objectives, directed the searches, and revised the mathematical presentation. The accompanying Lean certificate checks the formal statements and their axiom dependencies; it does not certify the prose exposition, literature comparison, or claims of originality. The author is responsible for the contents of the paper.

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